

```

> restart;
> with(LinearAlgebra);
[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix,
BidiagonalForm, BilinearForm, CARE, CharacteristicMatrix,
CharacteristicPolynomial, Column, ColumnDimension, ColumnOperation,
ColumnSpace, CompanionMatrix, CompressedSparseForm,
ConditionNumber, ConstantMatrix, ConstantVector, Copy,
CreatePermutation, CrossProduct, DARE, DeleteColumn, DeleteRow,
Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions,
DotProduct, EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal,
ForwardSubstitute, FrobeniusForm, FromCompressedSparseForm,
FromSplitForm, GaussianElimination, GenerateEquations,
GenerateMatrix, Generic, GetResultDataType, GetResultShape,
GivensRotationMatrix, GramSchmidt, HankelMatrix, HermiteForm,
HermitianTranspose, HessenbergForm, HilbertMatrix,
HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite,
IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm,
KroneckerProduct, LA_Main, LUDecomposition, LeastSquares,
LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential,
MatrixFunction, MatrixInverse, MatrixMatrixMultiply, MatrixNorm,
MatrixPower, MatrixScalarMultiply, MatrixVectorMultiply,
MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm,
Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm,
ProjectionMatrix, QRDecomposition, RandomMatrix, RandomVector, Rank,
RationalCanonicalForm, ReducedRowEchelonForm, Row, RowDimension,
RowOperation, RowSpace, ScalarMatrix, ScalarMultiply, ScalarVector,
SchurForm, SingularValues, SmithForm, SplitForm,
StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis,
SylvesterMatrix, SylvesterSolve, ToeplitzMatrix, Trace, Transpose,
TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd,
VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply,
ZeroMatrix, ZeroVector, Zip]

```

(1)

```

> with(linalg):with(Student[LinearAlgebra]): with (LinearAlgebra):
> # Exercise 1
> lambda[1]:=-2

```

$$\lambda_1 := -2 \quad (2)$$

```

> lambda[2]:=-1+I;

```

$$\lambda_2 := -1 + I \quad (3)$$

```
> lambda[3]:=-1-I;
```

$$\lambda_3 := -1 - I \quad (4)$$

```
> u[1]:=Matrix([[0],[0],[1]])
```

$$u_1 := \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (5)$$

```
> u[2]:=Matrix([[1+I],[1],[0]])
```

$$u_2 := \begin{bmatrix} 1 + I \\ 1 \\ 0 \end{bmatrix} \quad (6)$$

```
> u[3]:=Matrix([[1-I],[1],[0]])
```

$$u_3 := \begin{bmatrix} 1 - I \\ 1 \\ 0 \end{bmatrix} \quad (7)$$

```
> P:=Matrix([u[1],u[2],u[3]])
```

$$P := \begin{bmatrix} 0 & 1 + I & 1 - I \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (8)$$

```
> A:=Matrix([[0,-2,0],[1,-2,0],[0,0,-2]]);
```

$$A := \begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad (9)$$

```
> D1:=Matrix([[lambda[1],0,0],[0,lambda[2],0],[0,0,lambda[3]]]);
```

$$D1 := \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 + I & 0 \\ 0 & 0 & -1 - I \end{bmatrix} \quad (10)$$

```
> # a) compute those matrices, deduce that uk is an eigenvector
corresponding to eigenvalue lambda[k]
> evalm(A &* u[1] - lambda[1]*u[1]);
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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```
> evalm(A &* u[1] - lambda[2]*u[1]);
```

$$\begin{bmatrix} 0 \\ 0 \\ -1 - I \end{bmatrix}$$

(12)

```
> evalm(A &* u[2] - lambda[2]*u[2]);
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(13)

```
> evalm(A &* u[2] - lambda[3]*u[2]);
```

$$\begin{bmatrix} -2 + 2I \\ 2I \\ 0 \end{bmatrix}$$

(14)

```
> evalm(A &* u[3] - lambda[3]*u[3]);
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(15)

```
> simplify(evalm(A &* u[1] - lambda[1]*u[1]))
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(16)

```
> simplify(evalm(A &* u[2] - lambda[2]*u[2]))
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(17)

```
> simplify(evalm(A &* u[3] - lambda[3]*u[3]))
```

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(18)

```
> Eigenvalues(A)
```

$$\text{Eigenvalues} \left(\begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \right)$$

(19)

```
> eigenvalues(A)
```

$$\text{eigenvalues} \left(\begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \right)$$

(20)

```
> eigenvectors(A)
```

$$\text{eigenvectors} \left(\begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \right)$$

(21)

```
> Eigenvectors(A)
```

$$\text{Eigenvectors} \left(\begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \right)$$

(22)

```
> # because A*u[k]=lambda[k]*u[k], k=1,2,3 => u[k] is an
eigenvector corresponding to the eigenvalue lambda[k], for each
k=1,2,3
```

```
> # b) compute a-pdp^-1 and deduce that a is diagonalizable over C
```

```
> B:=evalm(P &* D1 &* P^(-1));
```

$$B := \begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

(23)

```
> B:=LinearAlgebra:-Multiply(P,D1,MatrixInverse(P));
```

Error. (in LinearAlgebra:-Multiply) invalid output option:
 MatrixInverse(Matrix(3, 3, [[0.1+I,1-I],[0.1,1],[1,0,0]]))

> P:=<u[1] | u[2] | u[3]>;

$$P := \begin{bmatrix} 0 & 1+I & 1-I \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (24)$$

> B:=LinearAlgebra:-Multiply(P,D1,LinearAlgebra:-MatrixInverse(P));
 Error. (in LinearAlgebra:-Multiply) invalid output option: Matrix(3, 3, [[0,0,1],[-1/2*I,1/2+1/2*I,0],[1/2*I,1/2-1/2*I,0]])

> P:=Matrix([[u[1][1,1],u[2][1,1],u[3][1,1]],[u[1][2,1],u[2][2,1],u[3][2,1]],[u[1][3,1],u[2][3,1],u[3][3,1]]]);

$$P := \begin{bmatrix} 0 & 1+I & 1-I \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (25)$$

> P:=Matrix([u[1],u[2],u[3]])

$$P := \begin{bmatrix} 0 & 1+I & 1-I \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (26)$$

> u[1]:=Matrix([[0],[0],[1]])

$$u_1 := \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (27)$$

> u[2]:=Matrix([[1+I],[1],[0]])

$$u_2 := \begin{bmatrix} 1+I \\ 1 \\ 0 \end{bmatrix} \quad (28)$$

> u[3]:=Matrix([[1-I],[1],[0]])

$$u_3 := \begin{bmatrix} 1-I \\ 1 \\ 0 \end{bmatrix} \quad (29)$$

```
> P:=Matrix([u[1],u[2],u[3]])
```

$$P := \begin{bmatrix} 0 & 1+I & 1-I \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (30)$$

```
> B:=evalm(P &* D1 &* P^(-1));
```

$$B := \begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad (31)$$

```
> lambda[1]:=-2
```

$$\lambda_1 := -2 \quad (32)$$

```
> lambda[2]:=-1+I;
```

$$\lambda_2 := -1 + I \quad (33)$$

```
> lambda[3]:=-1-I;
```

$$\lambda_3 := -1 - I \quad (34)$$

```
> D1:=Matrix([[lambda[1],0,0],[0,lambda[2],0],[0,0,lambda[3]]]);
```

$$D1 := \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1+I & 0 \\ 0 & 0 & -1-I \end{bmatrix} \quad (35)$$

```
> B:=evalm(P &* D1 &* P^(-1));
```

$$B := \begin{bmatrix} 0 & -2 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad (36)$$

```
> evalm(A-B)
```

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (37)$$

```
> # c)
```

```
> lambda[1]
```

$$-2 \quad (38)$$

$$\begin{aligned} &> \text{lambda}[2] \\ & \qquad \qquad \qquad -1 + I \end{aligned} \tag{39}$$

$$\begin{aligned} &> \text{lambda}[3] \\ & \qquad \qquad \qquad -1 - I \end{aligned} \tag{40}$$

$$\begin{aligned} &> D1 \\ & \qquad \qquad \qquad \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 + I & 0 \\ 0 & 0 & -1 - I \end{bmatrix} \end{aligned} \tag{41}$$

$$\begin{aligned} &> \text{LinearAlgebra:-MatrixExponential}(t*D1) \\ & \qquad \qquad \qquad \begin{bmatrix} e^{-2t} & 0 & 0 \\ 0 & e^{-t} \cos(t) + I e^{-t} \sin(t) & 0 \\ 0 & 0 & e^{-t} \cos(t) - I e^{-t} \sin(t) \end{bmatrix} \end{aligned} \tag{42}$$

$$\begin{aligned} &> ee := \text{LinearAlgebra:-MatrixExponential}(t*A); \\ ee := & \begin{bmatrix} e^{-t} \cos(t) + e^{-t} \sin(t) & -2 e^{-t} \sin(t) & 0 \\ e^{-t} \sin(t) & e^{-t} \cos(t) - e^{-t} \sin(t) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \end{aligned} \tag{43}$$

$$\begin{aligned} &> \text{simplify}(ee) \\ & \qquad \qquad \qquad \begin{bmatrix} e^{-t} (\cos(t) + \sin(t)) & -2 e^{-t} \sin(t) & 0 \\ e^{-t} \sin(t) & e^{-t} (\cos(t) - \sin(t)) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \end{aligned} \tag{44}$$

$$\begin{aligned} &> \text{combine}(\%) \\ & \qquad \qquad \qquad \begin{bmatrix} e^{-t} (\cos(t) + \sin(t)) & -2 e^{-t} \sin(t) & 0 \\ e^{-t} \sin(t) & e^{-t} (\cos(t) - \sin(t)) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \end{aligned} \tag{45}$$

> simplify(%)

$$\begin{bmatrix} e^{-t}(\cos(t) + \sin(t)) & -2e^{-t}\sin(t) & 0 \\ e^{-t}\sin(t) & e^{-t}(\cos(t) - \sin(t)) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \quad (46)$$

```
> combine(%%)
```

$$\begin{bmatrix} e^{-t}(\cos(t) + \sin(t)) & -2e^{-t}\sin(t) & 0 \\ e^{-t}\sin(t) & e^{-t}(\cos(t) - \sin(t)) & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix} \quad (47)$$

```
> # d)
```

```
> limit(ee, t=infinity)
```

Error, invalid input: limit expects its 1st argument, expr, to be of type Or(equation,algebraic), but received Matrix(3, 3, [[exp(-t)*cos(t)+exp(-t)*sin(t),-2*exp(-t)*sin(t),0],[exp(-t)*sin(t),exp(-t)*cos(t)-exp(-t)*sin(t),0],[0,0,exp(-2*t)])])

```
> map(x->limit(x,t=infinity),ee)
```

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (48)$$

```
> for i from 1 to 3 do
  for j from 1 to 3 do
    lim:=limit(ee[i][j],t=infinity);
    print(lim)
  end do:
end do:
```

```
0
0
0
0
0
0
0
0
0
0
```

(49)

```
[> # e)
```

```
[> ec1:=diff(x[1](t),t)=-2*x[2](t)
```

$$ec1 := \frac{d}{dt} x_1(t) = -2 x_2(t) \quad (50)$$

```
[> ec2:=diff(x[2](t),t)=x[1](t)-2*x[2](t)
```

$$ec2 := \frac{d}{dt} x_2(t) = x_1(t) - 2 x_2(t) \quad (51)$$

```
[> ec3:=diff(x[3](t),t)=-2*x[3](t)
```

$$ec3 := \frac{d}{dt} x_3(t) = -2 x_3(t) \quad (52)$$

```
[> sist:=ec1,ec2,ec3;
```

$$sist := \frac{d}{dt} x_1(t) = -2 x_2(t), \frac{d}{dt} x_2(t) = x_1(t) - 2 x_2(t), \frac{d}{dt} x_3(t) = -2 x_3(t) \quad (53)$$

```
[> sol:=dsolve({sist},{x[1],x[2],x[3]});
```

$$sol := \left\{ x_1(t) = e^{-t} (\sin(t) c_1 + \cos(t) c_2), x_2(t) \right. \\ \left. = \frac{e^{-t} (\sin(t) c_1 + \sin(t) c_2 - \cos(t) c_1 + \cos(t) c_2)}{2}, x_3(t) = c_3 e^{-2t} \right\} \quad (54)$$

```
[> rhs(sol[1]);
```

$$e^{-t} (\sin(t) c_1 + \cos(t) c_2) \quad (55)$$

```
[> limit(rhs(sol[1]),t=infinity);
```

$$0 \quad (56)$$

```
[> limit(rhs(sol[2]),t=infinity);
```

$$0 \quad (57)$$

```
[> limit(rhs(sol[3]),t=infinity);
```

$$0 \quad (58)$$

```
[> # Exercise 2
```

```
[> restart;
```

```
[> with(LinearAlgebra)
```

[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix, (59)
BidiagonalForm, BilinearForm, CARE, CharacteristicMatrix,
CharacteristicPolynomial, Column, ColumnDimension, ColumnOperation,
ColumnSpace, CompanionMatrix, CompressedSparseForm,
ConditionNumber, ConstantMatrix, ConstantVector, Copy,
CreatePermutation, CrossProduct, DARE, DeleteColumn, DeleteRow,
Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions,
DotProduct, EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal,
ForwardSubstitute, FrobeniusForm, FromCompressedSparseForm,

FromSplitForm, GaussianElimination, GenerateEquations, GenerateMatrix, Generic, GetResultDataType, GetResultShape, GivensRotationMatrix, GramSchmidt, HankelMatrix, HermiteForm, HermitianTranspose, HessenbergForm, HilbertMatrix, HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite, IsOrthogonal, IsSimilar, IsUnitary, JordanBlockMatrix, JordanForm, KroneckerProduct, LA_Main, LUDecomposition, LeastSquares, LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential, MatrixFunction, MatrixInverse, MatrixMatrixMultiply, MatrixNorm, MatrixPower, MatrixScalarMultiply, MatrixVectorMultiply, MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm, Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm, ProjectionMatrix, QRDecomposition, RandomMatrix, RandomVector, Rank, RationalCanonicalForm, ReducedRowEchelonForm, Row, RowDimension, RowOperation, RowSpace, ScalarMatrix, ScalarMultiply, ScalarVector, SchurForm, SingularValues, SmithForm, SplitForm, StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis, SylvesterMatrix, SylvesterSolve, ToeplitzMatrix, Trace, Transpose, TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd, VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply, ZeroMatrix, ZeroVector, Zip]

> D1:=LinearAlgebra:-DiagonalMatrix([2,2,-1,0]);

$$D1 := \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(60)

> P:=LinearAlgebra:-RandomMatrix(4,generator=-1..1);

$$P := \begin{bmatrix} -1 & 0 & 1 & 1 \\ -1 & 0 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

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> A:=simplify(P.D1.P^(-1));

$$A := \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \quad (62)$$

> eigenvals(A)

$$\text{eigenvals} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \right) \quad (63)$$

> eigenvalues(A)

$$\text{eigenvalues} \left(\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & -2 \\ \frac{3}{2} & -\frac{3}{2} & 0 & -2 \\ -\frac{3}{2} & -\frac{1}{2} & 2 & -2 \\ -\frac{3}{2} & \frac{3}{2} & 0 & 2 \end{bmatrix} \right) \quad (64)$$

> Eigenvalues(A);

(65)

$$\begin{bmatrix} 0 \\ -1 \\ 2 \\ 2 \end{bmatrix}$$

(65)

> Eigenvectors(A)

$$\begin{bmatrix} 0 \\ -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 & -1 & 0 \\ 1 & -1 & -1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

(66)

> # b)

> Determinant(A)

0

(67)

> CharacteristicPolynomial(A,lambda)

$$\lambda^4 - 3\lambda^3 + 4\lambda$$

(68)

> solve(lambda^4 - 3*lambda^3 + 4*lambda=0,lambda);

0, -1, 2, 2

(69)

> Eigenvectors(A);

$$\begin{bmatrix} 0 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -1 & 1 \\ 1 & 0 & -1 & -1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(70)

> JordanForm(A);

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

(71)

> MatrixInverse(A);

Error. (in MatrixInverse) singular matrix

> # METHOD GAUSS-JORDAN: zeroes both under and above main diagonal
- used for big stuff

> # Exercise 3